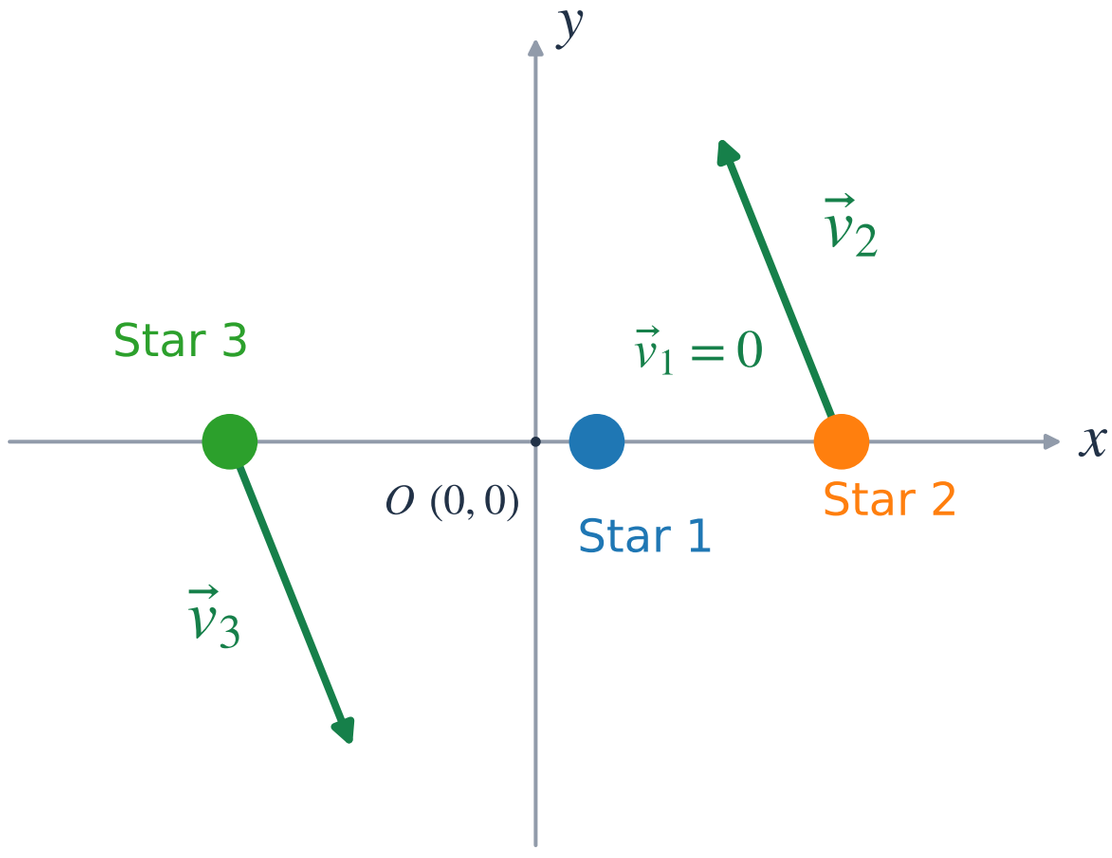
Initial positions and velocities of three stars



Star 1: initially at rest

$\vec{r}_1 = (0.1,\ 0)$

$\vec{v}_1 = (0,\ 0)$

Star 2

$\vec{r}_2 = (0.5,\ 0)$

$\vec{v}_2 = (-0.2,\ 0.5)$

Star 3

$\vec{r}_3 = (-0.5,\ 0)$

$\vec{v}_3 = (0.2,\ -0.5)$

$Gm_1 = Gm_2 = Gm_3 = 1$ | Green arrows show initial velocity, not force.